Selective Recovery for Multilingual ASR: Candidate Acquisition and Acceptance

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***Abstract*—Fallback recovery depends on both candidate acquisition and acceptance. We separate these decisions using a text-compression trigger and fixed four-feature selector. On 7,500 frozen Common Voice recordings, selective recovery lowers macro word error rate by 2.26 percentage points and matches unconditional recovery's error counts in every seed. Retrospective CV27 TEST comparisons retain 99.98% of positive edit gain. On matched banks, unweighted ranking trails expected-risk training across seeds, whereas advantages from direct adoption and edit-gap weighting depend on source and checkpoint. Timing across three checkpoints estimates up to 5.15% savings over unconditional reranked recovery, although Native remains faster. Code is available at https://github.com/Zhifan-Pan/ISCAS2027.**

***Index Terms*—multilingual speech recognition, selective recovery, candidate acquisition, N-best reranking, confidence estimation, conditional computation**

I. Introduction

Additional decoding improves recognition only when useful candidates are accepted. We separate acquisition from acceptance to study this interaction in multilingual ASR.

Multilingual N-best reranking [1], [2], discriminative rescoring [3]–[5], voting [6] and confusion networks [7] exploit existing hypotheses. Generative correction expands banks [8]–[11], while risk-based pruning and ASR sampling allocate computation [12], [13]. Conditional computation uses recognizer cascades [14], selective passes [15], encoder proxies [16], selective correction [17] or language-specific distillation [18]. Confidence need not predict computational benefit [19], because acceptance determines which acquired hypotheses improve output.

With recognizer and language labels fixed, we compare acquisition within sources. A frozen panel evaluates the original policy, while retrospective controls assess acceptance objectives [20]–[22] and acquisition budgets.

II. Controlled Acquisition and Acceptance

A. Recording-level acquisition

Normalizing and deduplicating five beams per window i forms B_i, retaining first occurrences and sources. With raw top beam b_i and fallback g_i, the expanded bank $F_i = B_i \cup \{g_i\}$ contains at most six candidates. The outer $\mathrm{CR}_{\mathrm{text}}$ divides top-beam UTF-8 byte length by zlib-compressed length, assigning zero to empty bytes. With $z_i = \mathbf{1}\{\mathrm{CR}_{\mathrm{text},i} > 2.4\}$ and $z^{\mathrm{rec}} = \max_i z_i$, any trigger acquires fallback for every window, including singletons.

Recovery follows Hugging Face temperatures $0, 0.2, \ldots, 1$ with compression, log-probability and no-speech thresholds 1.35, -1 and 0.6. Internal compression uses serialized token-ID bytes, unlike the outer text scale. Transformers v4.57.6 lists 1.35 as common [23]. Native HF uses the same configuration, with conditional retries and only the final output entering the bank.

B. Four-feature Shared acceptance

For features $f(y) = (a, p, t, r)$, decoder score a averages teacher-forced log probabilities over T body tokens, excluding prefix, padding and end tokens, while $t = \log(1 + T)$. XLM-R-base pseudo-log-likelihood p [24], [25], or PLL, averages individually masked nonspecial subword log probabilities in centered contexts of at most 510 subwords of the original string, rather than within-word masking [26]. Missing body tokens give $a = t = 0$ and missing PLL targets give $p = 0$, whereas normalization alone need not produce zero features.

For normalized words $w_{1:M}$, the sequence-wide overlapping count is $c_y(u) = \sum_{k=1}^{M-n+1} \mathbf{1}\{w_{k:k+n-1} = u\}$. We measure repetition by

$$r_{n,h}(y) = \frac{\sum_{j=1}^{M-n+1} \mathbf{1}\{c_y(w_{j:j+n-1}) \geq h\}}{\max(1, M - n + 1)}. \tag{1}$$

We use $r = r_{4,3}$, which is zero for $M < 4$. Repetitive sequences also motivate unlikelihood training [27].

Centering $x_i(y) = f(y) - f(b_i)$ preserves rankings and softmax probabilities. Feature-wise RMS scales $s \in \mathbb{R}^4_{>0}$ use training-fold pairs $\{(i, k) : i \in \bigcup_{\ell=1}^{L} \mathcal{I}_\ell,\ y_{ik} \in C_i \setminus \{b_i\}\}$ in fitting banks C_i, replacing scales below 10^{-12} with one. With edit benefit $\Delta_{ik} = E(b_i, y_i^*) - E(y_{ik}, y_i^*)$ and $\pi_{ik} = \mathrm{softmax}_k\big(\theta^\top \mathrm{diag}(s)^{-1} x_{ik}\big)$, expected-risk fitting [20] minimizes

$$\mathcal{L}(\theta) = -\frac{1}{L} \sum_{\ell=1}^{L} \frac{\sum_{i \in \mathcal{I}_\ell} \sum_k \pi_{ik} \Delta_{ik}}{N_\ell} + \lambda \|\theta\|_2^2. \tag{2}$$

Here $\mathcal{I}_\ell = \{i : \mathrm{lang}(i) = \ell\}$ indexes fitting windows, N_ℓ counts reference words and $L = 5$.

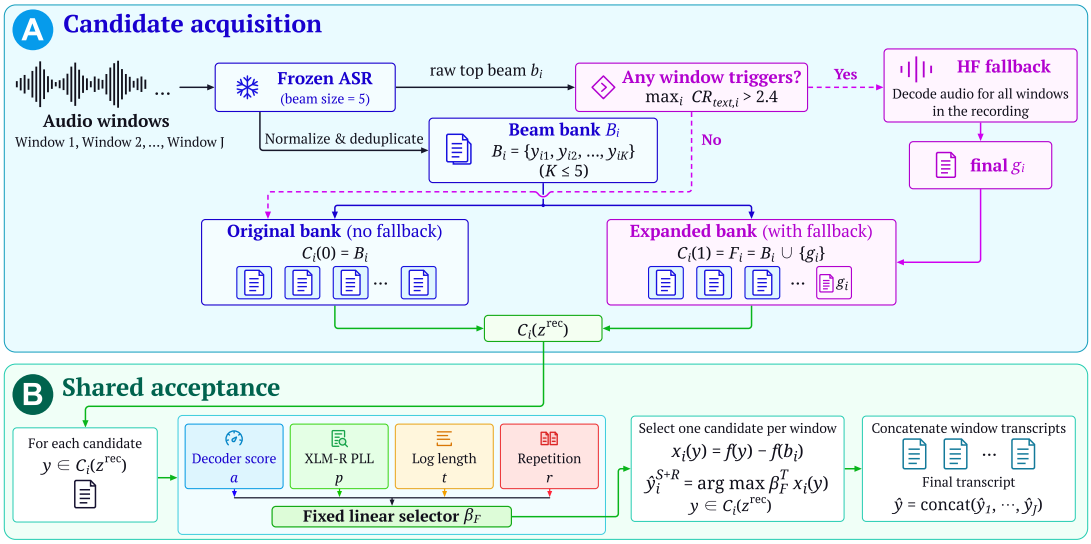


Fig. 1. Acquisition (A) expands every window's candidate bank if any window triggers. Acceptance (B) applies the same fixed four-feature selector to the original or expanded bank. Selected window transcripts are then concatenated.

Shared weights $\beta_F = \text{diag}(s)^{-1}\theta$ are fitted per run on greedy-expanded DEV and fixed across sources, resolving ties by bank order. With $\hat{y}_i^B = \arg\max_{y \in B_i} \beta_F^\top x_i(y)$, $C_i(0) = B_i$ and $C_i(1) = F_i$, the policies are

$$\hat{y}_i^{S+R} = \arg \max_{y \in C_i(z^{\text{rec}})} \beta_F^\top x_i(y),$$
$$\hat{y}_i^{\text{Direct}} = \begin{cases} \hat{y}_i^B, & z^{\text{rec}} = 0, \\ g_i, & z^{\text{rec}} = 1. \end{cases} \tag{3}$$

B reranks B_i, whereas N+R requests and reranks F_i unconditionally. Direct bypasses selection and PLL on triggers, retaining B elsewhere. CV27 has $z^{\text{rec}} = z_i$, while multi-window FLEURS uses the recording flag. Window transcripts are concatenated for full-reference scoring, with Shared fixing acceptance to isolate acquisition.

C. Matched acceptance controls

G4 uses the original β_F. H4 refits expected risk on complete HF-expanded DEV banks, while H3 uses (a, t, r). HR4 fits RankNet [21] on H4's banks and four features, favoring lower-edit candidates in unequal-edit pairs with language-reference normalization and equal total bank weight. HWR4 weights pairs by $|E_j - E_k|/c$, where c is the training-fold mean edit gap under the original pair weights.

Five folds group connected clients and reference texts within languages. Each training fold determines RMS scales, and held-fold macro WER selects among 49 log-spaced penalties from 10^{-4} to 1, favoring stronger regularization on ties. Full-DEV fitting uses zero-initialized L-BFGS-B. Singletons retain errors and reference denominators without contributing ranking pairs.

Minimum Bayes risk decoding [22] minimizes expected word edits on the same banks using uniform mass in U-MBR or body scores $\text{softmax}(Ta/\tau)$ in D-MBR. These scores are not calibrated sequence posteriors. DEV WER selects $\tau \in \{0.1, 0.3, 1, 3, 10\}$, preferring larger values on ties, while candidate ties retain bank order. A *complete policy* applies its own acceptor to both banks. A *triggered-subset evaluation* changes only triggered acceptance, preserving G4 elsewhere and full-panel denominators. H3Direct uses H3 elsewhere and final HF on triggers.

III. Data and Evaluation

We adapt Whisper-small [29] on 44.60 h of FLEURS TRAIN [28] to Arabic, Persian, Polish, Turkish and Italian using LoRA [30] with three different random seeds. Each run uses 4,656 updates, rank and scale four, zero dropout and all attention projections. Adaptation excludes CV27 TRAIN, while selectors and gates use CV27 DEV rather than FLEURS DEV. Complete TEST contains 3,665 FLEURS and 58,514 Common Voice 27.0 recordings [31]. Its use during development makes TEST comparisons retrospective.

Confirmation uses 7,500 CV27 TRAIN recordings, 1,500 per language, excluding previously accessed recordings, audio and reference texts. Predictions, stopping rules and the one-sided 0.10 percentage-point (pp) margin were locked before reference evaluation under procedural controls. Within-panel correlation is addressed by 10,000 synchronized bootstrap draws of speaker components within languages. Recurring speakers across splits restrict generalization to the observed speaker-clustered sampling frame, while later gates, refits and window policies remain retrospective.

Decoding uses known languages, bfloat16, five beams, length penalty one, native suppression and a 444-token cap in Transformers 4.57.6. At 16 kHz, N samples give $J = \lceil N/480000 \rceil$ windows bounded by $\lfloor jN/J \rfloor$ without previous-window text conditioning. Deduplication, WER and r share NFC normalization, digit unification, punctuation-to-space, Persian mapping and Turkish casing [32], with fixed source RNG mappings.

WER aggregates within languages before averaging five languages and three seeds, whereas calls pool seed–recordings. TEST intervals use 2,000 paired draws of client/reference components within languages, synchronized across seeds and conditional on frozen fits and source draws. FLEURS comparisons are descriptive because component metadata are unverified.

Costs use 255 FLEURS and 472 CV27 recordings, covering first-checkpoint triggers or Native-output changes plus ten ordinary recordings per language and duration quintile, weighted by inverse inclusion probability. Five FLEURS and 207 CV27 later-checkpoint triggers are unsampled, or 0.05% and 0.12% of seed–recordings in strata with nonzero inclusion probabilities. Triggered N+R and S+R perform identical acquisition and scoring (zero paired algorithmic difference). Study A's relative savings (5.15% and 4.57%) closely match Study B's single-checkpoint complete census (5.18% and 4.54%). Both studies use three randomized paired A40 rounds, resident models, encoder/first-search reuse and 32-mask PLL batches. Waveform-to-output timing excludes loading, warmup and interfered blocks. Session-conditional intervals use 4,000 paired stratified bootstrap draws with finite-population correction and synchronized, equally weighted checkpoints in A.

IV. Results

A. Acquisition with fixed acceptance

TABLE I
WER (%) WITH PP CONTRASTS COMPUTED BEFORE ROUNDING. CHECKS DENOTE HF ACQUISITION, CR GATING AND R RESELECTION, WITH † FOR R OFF-TRIGGER.

Path	HF	CR	R	Scorer	FLEURS	CV27
Raw beam-5 top-1	–	–	–	–	18.90	36.41
Native HF	✓	–	–	–	18.55	33.80
B	–	–	✓	G4	17.45	33.98
N+R	✓	–	✓	G4	17.09	31.68
S+R	✓	✓	✓	G4	17.09	31.68
Direct	✓	✓	✓†	G4	17.09	31.46
H3Direct	✓	✓	✓†	H3	18.68	33.64
Reference-assisted lower bounds on TEST						
Original-bank oracle					16.19	30.19
Full-HF oracle					15.83	27.68
CR-routed oracle					15.83	27.68
S+R gap to routed oracle, pp					1.26	4.00

Frozen CV27 confirmation with 7,500 recordings			
	B	N+R = S+R	Gain
Macro-average	35.02	32.75	2.26
Contrast	ΔWER		95% upper bound
S+R − B	-2.26		-1.32
S+R − N+R	0.00		0.00

On the frozen panel, S+R gains 2.26 pp over B, ranging from 0.20 to 4.54 pp across seeds. Substitution, deletion and insertion counts match N+R in every seed, meeting the prespecified 0.10 pp margin within this panel rather than guaranteeing zero loss under distribution shift. Among 61 triggers, 49 improve edits over B and two worsen them.

TABLE II
SAME-HF TEST WER (%) AND GAIN B−S (PP) BEFORE ROUNDING. CHECKS DENOTE PLL, PAIRWISE P AND EDIT-GAP WEIGHTING W.

Acceptor	PLL	P	W	FLEURS			CV27		
				B	S	Gain	B	S	Gain
G4	✓	–	–	17.45	17.09	0.37	33.98	31.68	2.30
H4	✓	–	–	17.46	17.09	0.37	33.97	31.52	2.45
HR4	✓	✓	–	17.51	17.14	0.37	34.02	32.30	1.72
H3	–	–	–	19.05	18.68	0.37	36.15	33.77	2.38
HWR4	✓	✓	✓	17.47	17.10	0.37	33.97	31.44	2.53
U-MBR	–	–	–	19.14	19.14	0.00	36.91	36.88	0.03
D-MBR	–	–	–	18.92	18.75	0.17	36.09	34.48	1.60

Original family CV27 ΔWER [individual 97.5% CI]		
Triggered HR4 − H4	+0.72	[+0.40, +1.16]
Complete H3 − H4	+2.25	[+2.16, +2.34]
Later two-comparison family		
Triggered U-MBR − H4	+2.51	[+1.93, +3.18]
Triggered HWR4 − H4	-0.08	[-0.15, -0.03]

Retrospective Direct reaches 32.71% WER against locked S+R's 32.75%, both beating B and Native's 34.87% by over 2.1 pp. Their difference before rounding is -0.05 pp with a paired speaker-bootstrap 95% interval from -0.19 to $+0.04$ pp.

The archive froze G4 scores and transcripts to isolate acquisition, whereas refitted acceptors use full TEST.

On TEST, FLEURS retains all ten beneficial instances at 0.09% calls, while CV27 retains 370 of 373 at 0.30%, missing five edits. Normalized N+R and S+R outputs differ on one FLEURS and 13 CV27 nontriggers despite nearly identical WER. Reference-assisted oracles minimize edits over original, expanded or CR-routed banks, including concatenated two-window candidates. Routing changes this bound by under 0.01 pp, whereas CV27 S+R remains 4.00 pp above it. Raw top-1 to B measures reselection rather than PLL alone.

B. Acceptance changes recovery opportunities

Direct lowers CV27 WER by bypassing G4 on triggers despite retaining only 67.13% of its own positive edit gain, against Shared's 99.98%, using full-language denominators. Relative to unconditional final HF, it forgoes 1.24 pp of beneficial changes but avoids 3.58 pp of harmful changes, so positive gain alone does not determine accuracy.

Shared beats Direct by 0.53 pp with greedy recovery and 0.43 pp with one $T = 0.2$ draw. HF reverses the ranking by 0.22 pp, with a 95% interval from 0.11 to 0.37 pp. Only one checkpoint improves (-0.68 pp versus $+0.00$ and $+0.02$ pp). DEV favored HF reselection at 30.08% versus 30.09%, motivating fixed acceptance.

C. Acceptance on matched HF banks

HWR4 achieves the lowest complete CV27 WER of 31.44% with the same four-feature architecture (Table II). Triggered HR4 trails H4 in every seed ($+0.91$, $+0.08$ and $+1.17$ pp), whereas edit-gap weighting reverses the mean contrast to -0.08 pp through a 0.25 pp improvement in one checkpoint. Intervals provide 95% Bonferroni coverage per family, using 10,000 synchronized draws for the later family.

TABLE III
POPULATION-WEIGHTED MS AND OBSERVED PEAK ALLOCATED MIB WITH BOTH MODELS RESIDENT. EACH STUDY USES THREE PAIRED ROUNDS.

Path	FLEURS		CV27	
	ms	MiB	ms	MiB
A. Four paths, three checkpoints				
Native HF	1291.7	2980.8	517.4	2813.8
B	1352.7	3144.9	591.9	3133.1
N+R	1436.8	2988.3	639.2	3133.1
S+R	1362.8	3189.7	610.0	3133.1
B. Seven paths, one checkpoint				
Native HF	1304.8	2978.4	531.2	2811.8
Raw top-1	1222.6	2946.6	479.9	2571.0
B	1364.0	3142.4	605.1	3131.1
N+R	1449.6	2986.3	657.9	3131.1
S+R	1374.5	3187.2	628.0	3131.1
Direct	1372.8	3187.2	625.8	2811.8
H3Direct	1314.0	3187.2	586.4	2811.8
S+R saving versus N+R in % [pointwise 95% CI]				
A	5.15 [2.84, 9.41]		4.57 [4.40, 4.73]	
B	5.18 [2.88, 9.39]		4.54 [4.44, 4.66]	
Within-study B strategy differences, not stage times				
B − Raw top-1, ms	141.4		125.2	
N+R − B, ms	85.6		52.8	
S+R − B, ms	10.5		22.9	
Increment saved	87.7%		56.6%	

FLEURS gains all round to 0.37 pp because S equals its own B off-trigger, while H3's B has two fewer errors on one of ten triggers and S errors match. Language and checkpoint averaging reduces this difference below 0.006 pp. Nontriggers account for 97.23% of the refitted H3–H4 WER gap, measuring an output partition rather than PLL causality or time share.

D. Cost, heterogeneity and supporting diagnostics

For systems using B, S+R preserves N+R's accuracy gain while reducing recovery overhead, although Native and B remain faster (Table III). Study B's reranking increments over raw top-1 are 141.4 ms on FLEURS and 125.2 ms on CV27, while S+R saves 87.7% and 56.6% of N+R's increment over B. The unweighted CV27 profile averages 85.9 ms for teacher scoring and 149.3 ms for PLL over 25 recordings, 23 with multiple candidates. Population estimates include more singletons, which skip both stages. Raw top-1 and H3Direct omit PLL. H3Direct saves 6.29% against Direct but takes 55.2 ms longer than Native on CV27, with a 95% interval from 52.3 to 58.2 ms, while FLEURS' interval spans zero. H4 and HWR4 update only G4's four weights in the same scoring pipeline. Selectively skipping PLL on non-triggered windows remains a future optimization.

Empty TEST candidates enter only through HF fallback, giving $f(\emptyset) = (0,0,0,0)$ and centered score $\beta_F^\top x_i(\emptyset) = -\beta_F^\top f(b_i)$. Under signs $(+,+,-,-)$, the margin $-\beta_F^\top f(y) > 0$ favors emptiness when $a, p \le 0$, $t > 0$ and $r \ge 0$, even at $r = 0$. This ranks candidates rather than assigning transcript probabilities. Reverting all 18 empty acceptances in 11 CV27 recordings to B retains 91.52% of gain, or 2.10 pp at 31.88% WER. The counts suggest hallucination suppression [33], but acoustic attribution requires silence labels (Table IV). H4+e adds $e(y) = \mathbf{1}\{y = \emptyset\}$ on H4's DEV folds and grid. With one or two DEV empties per checkpoint, it matches H4's rounded 31.52% WER and retains all 18 acceptances.

TABLE IV
TEST GAINS AVERAGED OVER SEEDS. CALLS POOL SEED–RECORDINGS.

Language	FLEURS		CV27	
	Gain, pp	Calls, %	Gain, pp	Calls, %
Arabic	0.64	0.23	1.40	0.11
Persian	1.20	0.27	9.05	1.15
Polish	0.00	0.00	0.30	0.33
Turkish	0.00	0.00	0.51	0.03
Italian	0.00	0.00	0.23	0.02
Empty-HF acceptance of 18 instances in 11 CV27 recordings				
Acceptance minus B reversion	$S = -118$	$D = +120$	$I = -1995$	

Persian contributes 78.77% of CV27 TEST gain and 84.07% of confirmation gain, whereas Polish, Turkish and Italian have no FLEURS triggers or gains (Table IV). Script, noise and LoRA effects remain coupled. FLEURS' 22 two-window recordings yield 52.69% of net gain. Local Shared triggering on cached banks preserves counts while reducing window requests from 15 to 10 across full TEST. Local Direct changes WER by under 0.01 pp, though live RNG may differ. Transfer is mixed, with Bulgarian and Croatian greedy WER unchanged at 37.51% and S+R minus B of -0.08 pp on the earlier 4,431-recording panel, whose 95% interval runs from -0.23 to 0.00 pp. SeamlessM4T [34], [35] greedy CV27 WER falls from 12.96% to 12.69% at 0.08% calls. DEV-tuned MBR reaches 12.55% against risk's 12.66%, while full HF transfer remains unvalidated.

HF12 predicts fixed-G4 edit gain using ridge regression on grouped CV27 DEV folds. Its twelve features are $\log(1+d)$ for duration d in seconds, $\log(1+|B_i|)$, top-two decoder-score gap, softmax-a entropy, population standard deviation and range of a, mean $|a - a(b_i)|$, ranges of t and r, top-beam $\mathrm{CR}_{\mathrm{text}}$ and $r(b_i)$, and maximum repetition. Conditional on G4, predicted negative gain below DEV's $q = 3 \times 10^{-3}$ quantile triggers multi-candidate windows. HF12 and matched CR reach 31.71% CV27 WER at 0.29% and 0.26% calls. Their interval spans zero without establishing equivalence, although equal-call ranking favors HF12. FLEURS calls are 178, two and five across seeds.

V. CONCLUSION

Sparse acquisition matches unconditional recovery across three frozen-panel seeds and reduces latency within reranked pipelines. Expected-risk fitting consistently beats unweighted ranking, whereas Direct and edit-gap weighting depend on source and checkpoint. Separating these decisions supports selective acquisition under a fixed scorer while showing why acceptance and fallback generation require joint evaluation.

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